Diseases of the Upper GI System

NURA 340 Pathophysiology | Final Exam Lecture Study Guide

Esophagus, stomach, and duodenum

Lecture roadmap: Normal esophageal function → GERD → Esophagitis → Normal stomach function and protective defenses → Gastritis → Peptic ulcer disease → Upper GI bleeding (ulcer vs. variceal) → Wrap-up

How the instructor wants you to think: For every disorder, ask **what normal function has been disrupted**. Is something moving in the wrong direction? Has the protective lining been damaged? Has the injury gone deep enough to cause an ulcer or bleeding? Explain *why* each finding happens instead of memorizing lists. Nursing interventions, test ordering, and drug details are **not** tested in this course. The pharmacology mentions are only there to connect to your Pharm class.

How to read this guide: Everything without a tag comes from the lecture slides and transcript. Anything else is tagged: ADDED information not in the lecture; CLARIFIED a lecture point explained further or a garbled transcript line cleaned up; CORRECTED an error in the slides or transcript. Every tag is listed in the notes at the end.

Normal Upper GI Function

Esophagus

- A muscular tube that moves food and fluid from the mouth to the stomach by **peristalsis** (coordinated contractions).
- The **lower esophageal sphincter (LES)** relaxes during swallowing to let food in, then closes to keep stomach contents from moving back up. The **diaphragm** helps support this barrier.
- The esophageal lining has **much less protection against acid** than the stomach does.
- Small amounts of reflux are normal. **Saliva** (neutralizes acid) and **peristalsis** (pushes contents back down) clear them.
- Problems start when reflux is **frequent or prolonged**, or when the esophagus **can't clear it** effectively.
- Esophageal disorders either block the passage of food and liquid, or the structure and nerves/muscles stop working together. Swallowing becomes difficult or painful, and the whole digestive process can be affected.

Stomach

- Muscular walls mix and churn food and gradually move it into the **duodenum** (first part of the small intestine).
- **Parietal cells** make **hydrochloric acid (HCl)** and **intrinsic factor**. Intrinsic factor is needed to absorb **vitamin B12** in the **terminal ileum**.
- **Chief cells** release **pepsinogen**. Acid converts it to **pepsin**, which digests protein.
- Because acid and pepsin can digest the stomach's own tissue (muscle is protein), the stomach protects itself with:
  - a **mucus layer** that holds **bicarbonate** near the cell surface to neutralize acid
  - **tight junctions** between cells
  - **good blood flow** and ongoing tissue repair
  - **prostaglandins**, which support mucus, bicarbonate, and blood flow

Key idea: Stomach injury is a **balance** problem. Injury can happen when the **defenses weaken**, even if acid production is normal.

Gastroesophageal Reflux Disease (GERD)

GER vs. GERD: **GER** (gastroesophageal reflux) just means stomach contents moved backward into the esophagus. It happens occasionally, for example after a big meal, and is not a disease. **GERD** is when reflux causes **troublesome, recurrent symptoms or complications**.

Etiology	Stomach contents repeatedly flow back into the esophagus. Contributing factors: • **Increased abdominal pressure**: pregnancy, more adipose tissue around the abdomen or thorax • **Large meals** and **delayed stomach emptying** (contents stay longer, so more chance to reflux) • **Smoking** and **medications that lower LES pressure** • **Hiatal hernia**: changes the normal relationship between the stomach, LES, and diaphragm, weakening the barrier
Pathophysiology	The LES barrier doesn't work well. It may **relax when it should stay closed**. If reflux is frequent or clearance (saliva, peristalsis) is impaired, the esophageal lining is exposed to acid longer. Since it lacks the stomach's protection, this causes **irritation → inflammation → tissue injury**.

Manifestations	• **Heartburn**: burning behind the sternum. Worse **after eating** (fuller stomach), **bending** (more abdominal pressure), and **lying down** (loses gravity's help). • **Regurgitation**: stomach contents rise into the throat or mouth, often with a sour or acidic taste. • Cough, throat clearing, hoarseness, sore throat • **Dental erosion** if acid repeatedly reaches the mouth • These extra findings can have other causes, so interpret them with the overall symptom pattern.
Diagnosis	• A regular pattern of heartburn and regurgitation is often enough to start a **trial of treatment** (usually a PPI first). • **Red flags** that need more testing: difficulty swallowing, unexplained weight loss, signs of bleeding. • **Upper endoscopy** shows inflammation or damage. You don't need to know the grading system. • Not everyone with GERD has visible damage. If unclear, **reflux monitoring** measures how much acid reaches the esophagus.
Impact on health	• **Esophagitis** from repeated irritation • **Stricture**: scar tissue from healing narrows the esophagus and makes swallowing hard • **Barrett's esophagus**: long-term reflux changes the **type of cells** lining the lower esophagus. This raises cancer risk, although most people with Barrett's do not get cancer. Changes to cells increase cancer risk, which is why GERD is taken seriously. • **Interrupted sleep**; **aspiration** of refluxed contents into the lungs while lying down or asleep • Avoiding meals because eating hurts, which affects **nutrition** and quality of life
Pharm connection	**PPIs** do **not** stop the LES from relaxing or stop reflux. They **lower stomach acid**, so the refluxed contents cause less damage to the esophagus.

Discussion Question #1: Why do reflux symptoms worsen after a large meal or when lying down?
After a large meal, the stomach is **distended** with more pressure, which promotes **transient relaxation of the LES**. If the barrier is already impaired, contents move backward more easily. Lying down removes **gravity's help** in keeping contents in the stomach and clearing the esophagus. The link is **stomach distension + barrier function + how long the lining stays exposed**.

Esophagitis

Esophagitis = inflammation of the esophageal lining. GERD is the most common cause, but not the only one.

Etiology	• **GERD** (most common) • **Pill esophagitis**: a pill lodges in the esophagus and irritates the tissue as it dissolves • **Radiation** • **Infection** such as *Candida* (yeast), especially when the immune system is weakened • **Eosinophilic esophagitis (EoE)**: an abnormal immune response in the esophagus. Often seen in people with **food allergies, asthma, or eczema** (associated with, not the direct cause of every case).
Pathophysiology	Different causes, same result: **inflammation and injury** to tissue that needs to let food pass smoothly. • **Reflux-related:** repeated acid exposure damages the lining and triggers inflammation: increased blood flow, swelling, and stimulation of pain receptors. • **EoE: eosinophils** build up in the tissue and release substances that cause inflammation and damage (connects to the allergy content). • Ongoing injury and healing cause **scar tissue**, so the esophagus becomes less flexible and narrower (**stricture**).

Manifestations	• **Heartburn** (acid irritating the lining) • **Dysphagia** = **difficulty** swallowing; food feels slow or stuck • **Odynophagia** = **pain** with swallowing, because food passes over injured tissue • Swelling interferes with swallowing short-term; scarring causes longer-lasting narrowing • **Food impaction** (food becomes lodged)
Diagnosis	Must identify both the injury **and its cause**. **Upper endoscopy** shows inflammation, erosions, or narrowing. **Biopsy** helps tell the causes apart; for EoE it looks for **eosinophils in the esophageal tissue**. A **blood eosinophil count alone cannot confirm EoE**.
Impact on health	• Eating less or narrowing food choices, which affects **nutrition and weight** • **Stricture**: dense scar tissue that **does not relax** even after the inflammation improves, so swallowing can stay difficult • **Barrett's esophagus** is linked to **long-term reflux**, not to every cause of esophagitis
Management	Treatment must **match the cause**: reduce acid for reflux injury; address the immune response for EoE. Similar symptoms can come from different mechanisms, so the same medication won't always fix the cause.

Discussion Question #2: What changes in GERD-related esophagitis cause pain, difficulty swallowing, and heartburn?
Repeated acid exposure **injures the lining and triggers inflammation**. Irritation stimulates **sensory nerves**, causing the burning (heartburn). Swallowing over injured tissue hurts (**odynophagia**). **Swelling** makes food harder to move through (**dysphagia**). Repeated injury and healing cause **scar tissue** that keeps the opening narrow (**stricture**). Your answer should connect **acid exposure → inflammation → structural change → symptoms**.

Gastritis

Gastritis = inflammation of the stomach lining. The common thread is **disruption of the stomach's protective lining**.

Etiology	• ***Helicobacter pylori* (H. pylori)** infection: a very common cause • **Autoimmune gastritis**: the immune system attacks **parietal cells** • **NSAIDs** (e.g., ibuprofen), **alcohol**, **bile**. The term **gastropathy** may be used when tissue damage occurs with relatively little inflammation. • **Severe physiologic stress** (shock, critical illness) **reduces blood flow** to the lining and weakens its defenses. This means the stress of serious illness, not everyday emotional stress.
Pathophysiology	• **H. pylori** lives in the mucus layer and makes **urease**, which makes its immediate surroundings less acidic. This lets it survive and keep causing inflammation. • **NSAIDs reduce prostaglandin production**, so there is less mucus, less bicarbonate, and less mucosal blood flow. • Once protection is weaker, **acid and digestive enzymes** injure the stomach further. • **Autoimmune:** parietal cell damage also reduces **intrinsic factor**.
Manifestations	Burning or gnawing **epigastric** discomfort (middle upper abdomen), nausea, vomiting, **early fullness**, reduced intake and **weight loss**. Symptoms vary; some people have little discomfort despite inflammation.
Diagnosis	**Endoscopy ± biopsy** shows inflammation and tissue changes. For H. pylori: **urea breath test** or **stool antigen test** detects active infection. Finding the cause matters because it guides treatment.
Impact on health	**Bleeding** and **anemia** by two different pathways (see the box below).
Pharm connection	**PPIs** and **H2 blockers** reduce acid secretion so tissue can heal. **Antibiotics** are needed to eliminate H. pylori. Key point: **acid suppression helps healing but does not remove the cause.**

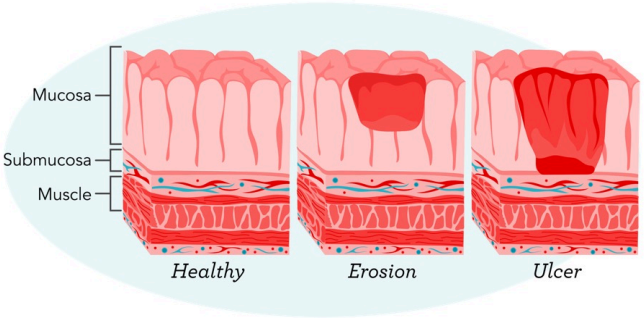
Discussion Question #5: How can gastritis lead to vitamin deficiency and anemia? Two pathways:

Pathway 1: Chronic blood loss	Pathway 2: Autoimmune damage to parietal cells
Damaged lining bleeds slowly over time → ongoing **iron** loss → body can't make enough hemoglobin → **iron deficiency anemia**	Parietal cells destroyed → less **intrinsic factor** → poor **vitamin B12** absorption in the terminal ileum → abnormal RBC development → **pernicious anemia** (B12 deficiency)

When you see gastritis and anemia together, ask: **is the person losing blood, or has the disease blocked something needed to make RBCs?** The second pathway starts in the stomach even though B12 is absorbed in the small intestine. Anemia is covered in more detail in a later module.

Peptic Ulcer Disease (PUD)

A **deeper, localized breakdown** of the lining of the **stomach or duodenum**. The key difference from gastritis or erosion is **depth**: an ulcer goes through the thin muscle layer of the mucosa (muscularis mucosae) into deeper tissue.



From the lecture slides: an erosion stays in the mucosa; an ulcer extends deeper.

Etiology	The two major causes are **H. pylori** and **NSAIDs**, both of which weaken mucosal protection. **Smoking** interferes with healing. **Spicy food is not a major cause**, though some foods may worsen discomfort.

Pathophysiology	• **H. pylori** → inflammation and disrupted defenses • **NSAIDs** → reduce the protective effects of **prostaglandins** • Either way, **acid and pepsin** damage the less-protected tissue. As the injury deepens it can reach **blood vessels** and cause bleeding, which is why ulcers can be more serious than discomfort.
Manifestations	• Burning or gnawing **epigastric pain**, nausea, bloating, fullness • Classic patterns: **gastric** ulcer pain worsens **after eating**; **duodenal** ulcer pain may **improve briefly with eating** and return later. These patterns are **not reliable enough to locate the ulcer**. • Some people have **little or no pain** and only learn they have an ulcer when **bleeding or another complication** develops.
Diagnosis	**Upper endoscopy** visualizes the ulcer. **Biopsy** can identify H. pylori or check abnormal tissue, especially in a gastric ulcer. A **CBC** can show anemia from blood loss but **doesn't show where** the bleeding came from or confirm an ulcer. Interpret everything together.
Impact on health	Complications follow the **depth and location** of the ulcer: • Reaches a blood vessel → **bleeding** • Goes through the full wall → **perforation** → contents leak into the abdominal cavity → **peritonitis** • Swelling or scarring near the stomach outlet → **obstruction** (contents can't leave the stomach) → persistent vomiting • Pain and reduced intake affect **nutrition**; bleeding and vomiting reduce **circulating volume**

Discussion Question #6: How can NSAIDs cause an ulcer when acid production is normal?

Prostaglandins maintain **mucus, bicarbonate, and mucosal blood flow**. NSAIDs **reduce prostaglandin production**, weakening those defenses. Now even a **normal amount of acid and pepsin** can injure the unprotected tissue. PUD develops because the **protective side of the balance** has weakened, not because acid increased.

Upper GI Bleeding

Upper GI = **esophagus, stomach, or duodenum**. Bleeding is a **complication of an underlying problem**, not a disease by itself.

Etiology	**Peptic ulcers** (important cause), **erosive gastritis**, **esophagitis**, **tears from forceful vomiting**, **ruptured varices**.
Pathophysiology	• **Ulcer or erosion:** tissue injury damages a blood vessel, and blood enters the GI tract. • **Variceal:** increased pressure in the **portal circulation** makes collateral veins in the esophagus or stomach enlarge. These dilated veins can **rupture and bleed heavily**. (Connects to cirrhosis and portal hypertension in the liver lecture.)
Manifestations	How blood looks depends on where it travels and how long it stays in the GI tract: • **Hematemesis** = vomiting blood. It may be **bright red**, or look like **coffee grounds** after exposure to stomach acid. Coffee-ground emesis means blood from the stomach. • **Melena** = **black, tarry stool**, from blood broken down as it passes through the intestines CLARIFIED • A **very rapid** upper GI bleed can produce **red or maroon blood from the rectum**. **Color alone doesn't establish the source.** • Weakness, dizziness, **tachycardia**, fainting: effects of blood loss on circulation and oxygen delivery
Diagnosis	Answer two questions: **Where is it coming from?** and **How much is it affecting the person?** • **Endoscopy** finds many upper GI sources. • **CBC** assesses blood loss but not the source. **Hemoglobin may stay near baseline early in an acute bleed**, because plasma and RBCs are lost together. It drops only after fluid shifts occur or fluids are given. • **BUN may rise**: blood in the GI tract is digested and absorbed, increasing the nitrogen load; reduced kidney perfusion adds to it. An elevated **BUN:creatinine ratio** is a **supporting clue, not proof**.
Impact on health	Depends on **how fast** the blood is lost: • **Slow, repeated bleeding** → iron stores depleted → **anemia** → less O_2 delivery → fatigue, low activity tolerance • **Rapid bleeding** → **volume problem**: less venous return → lower stroke volume → lower cardiac output. The body compensates first with **tachycardia and vasoconstriction**; continued loss → **hypotension and shock**.

Management	Treat the **source** of the bleeding **and** the **blood loss** itself. Some treatments are done through endoscopy (details are beyond this course).

Discussion Question #8: Ulcer-related vs. variceal bleeding
Ulcer: tissue damage goes deep enough to **erode into a blood vessel**. **Varices: increased portal pressure** enlarges veins until they **rupture**. Both cause serious blood loss, but they start by completely different mechanisms, which is why finding the source matters.

Quick Review

Disorder	Core mechanism	Hallmark findings	Key complications
GERD	LES barrier fails; esophagus exposed to acid it can't handle	Heartburn (worse after meals, bending, lying down), regurgitation	Esophagitis, stricture, Barrett's, aspiration
Esophagitis	Inflammation of the esophageal lining (GERD, pills, radiation, Candida, EoE)	Dysphagia, odynophagia, heartburn, food impaction	Stricture, poor nutrition
Gastritis	Stomach defenses disrupted (H. pylori, NSAIDs, autoimmune, alcohol, severe stress)	Epigastric burning, nausea, early fullness	Bleeding → iron deficiency anemia; autoimmune → pernicious (B12) anemia
PUD	Deeper breakdown of stomach or duodenal lining (H. pylori, NSAIDs)	Epigastric pain; may be silent until bleeding	Bleeding, perforation → peritonitis, outlet obstruction
Upper GI bleed	Ulcer/erosion into a vessel, or ruptured varices from portal HTN	Hematemesis (red or coffee-ground), melena, tachycardia, dizziness	Anemia (slow) or hypovolemic shock (rapid)

Instructor wrap-up: The esophagus must move food to the stomach and clear reflux. The stomach must digest food while protecting its own lining. When those functions break down, the **location and depth** of the injury explain the symptoms and complications. Practice explaining out loud: Why does inflammation make swallowing painful? Why can reduced mucosal protection lead to an ulcer? How can an ulcer affect circulation?

Self-Check ADDED

1. What is the difference between GER and GERD?
   GER is reflux itself, which can be normal and occasional. GERD is reflux causing troublesome recurrent symptoms or complications.
2. Why doesn't a PPI stop reflux?
   It doesn't change LES function. It lowers acid, so refluxed contents injure the esophagus less.
3. Dysphagia vs. odynophagia?
   Dysphagia = difficulty swallowing. Odynophagia = painful swallowing.
4. Why can't a blood eosinophil count confirm eosinophilic esophagitis?
   The diagnosis requires eosinophils in the esophageal tissue itself, found on biopsy.
5. How does H. pylori survive in the acidic stomach?
   It lives in the mucus layer and makes urease, which makes its immediate surroundings less acidic.
6. A patient with autoimmune gastritis is fatigued and pale. What kind of anemia should you suspect, and why?
   Pernicious (B12) anemia: parietal cell loss → no intrinsic factor → no B12 absorption.
7. What separates an ulcer from an erosion?
   Depth. An ulcer extends through the muscularis mucosae into deeper tissue.
8. A patient vomits material that looks like coffee grounds. What does that tell you?
   Blood that has been exposed to stomach acid: an upper GI (stomach) source.
9. Why might hemoglobin be normal right after a large GI bleed?
   Plasma and RBCs are lost together, so the concentration doesn't fall until fluid shifts happen or fluids are given.
10. Why can BUN rise in an upper GI bleed?
    Blood digested in the gut adds a nitrogen (protein) load, and reduced kidney perfusion contributes.
11. What are the early compensatory signs of rapid GI blood loss?
    Tachycardia and vasoconstriction. Later: hypotension and shock.

Notes on Sources, Additions & Clarifications

This guide was written from the lecture slides (Diseases of the Upper GI System, NURA 340) and the organized lecture transcript. The student slides contain only headings and images, so almost all content comes from the transcript.

- CLARIFIED **Melena (Upper GI Bleeding).** The transcript line reads as if *lower* GI bleeding produces black, tarry stool. In context, the lecturer is describing **upper** GI blood being broken down as it passes through the intestines, which is the standard cause of melena. The guide presents it that way.
- CLARIFIED **Spoken slips in the transcript** ("reflex" for reflux, "acetaminophen" corrected to ibuprofen as an NSAID example) are written in their corrected form.
- CLARIFIED **"Muscularis mucosae"** is the formal name for the "thin muscle layer within the mucosa" the lecturer describes for ulcer depth.
- ADDED **Self-Check questions** are my own, written from lecture content.
- Discussion question numbers (#1, #2, #5, #6, #8) match the slide numbering. The missing numbers are not in this deck.